Customer Service RAG Chatbot
Local AI Prototype

Case Study:
AI Engineer Intern
Customer Journey Analytics & Data Science

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Problem & Context

Digital Customer Channels in Automotive Industry

Customers interact via:

- Website
- Vehicle configurators
- Service portal

→ **Hundreds of recurring customer inquiries per day!**

Problem & Context

Problem: Customers repeatedly ask many questions like:

- What is included in the standard warranty?
- How often is service required?
- What is the battery warranty for EVs?
- How does the ordering process work?

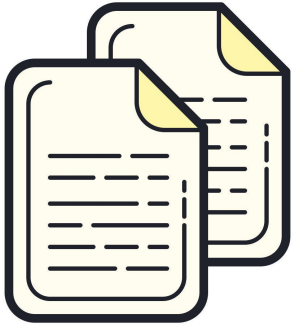
This leads to:

➔ High support workload

➔ Slow response times Inconsistent answers

➔ Poor customer experience

Problem & Context



Good News! → These answers already exist
But they are embedded in the documents.

→ **We need methods to process them…**

Possible Solution Approaches

- **FAQ Page:**
  - Simple
  - Controlled answers
  - Not scalable
  - Cannot handle paraphrasing
- **Traditional Search Engine**
  - Keyword-based
  - Fast
  - No conversational output
  - Can not handle phrased words
  - Requires manual reading

→ **Traditional approaches/search methods not enough**

We need something more efficient…

System

- **That understands natural language**
  - Handles variations in words (e.g. battery warranty and EV battery coverage)
  - Interprets questions the way customers actually ask them
- **Retrieves relevant knowledge**
  - Identifies the most relevant documents and their parts
  - Filters noise and irrelevant information
- **Generates precise answers**
  - Provides clear, explainable and structured responses
  - References the source of information

Beyond Simple Retrieval: NLP Techniques

Fine-Tuned LLM

Advantages

- Strong domain adaptation
- Deep understanding of company-specific terminology
- Fluent, natural responses
- Can internalize structured patterns across documents

Challenges

- Expensive to train and maintain
- Requires retraining when knowledge changes
- Hard to trace exact source of information
- Risk of outdated or memorized knowledge

RAG

Advantages

- No retraining required
- Dynamically retrieves relevant document chunks
- Easy to update
- More cost-efficient than full fine-tuning
- Enables source traceability and explainability
- Scales well as the knowledge base grows

Challenges

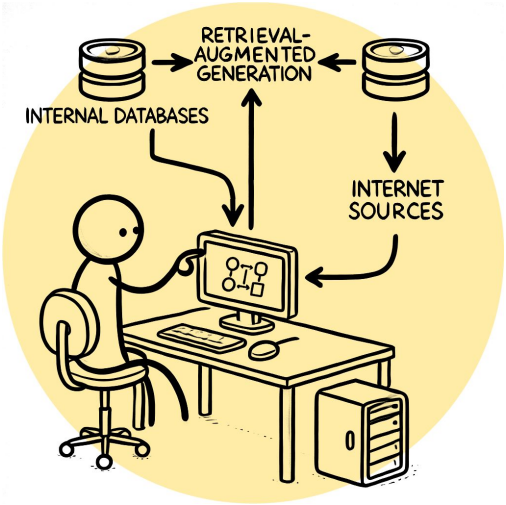
- Performance depends on retrieval quality
- Requires careful prompt engineering

Why RAG in This Context?

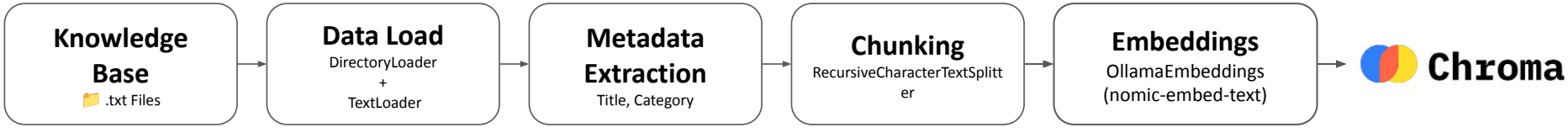
- Automotive documentation evolves continuously

  → static models are insufficient

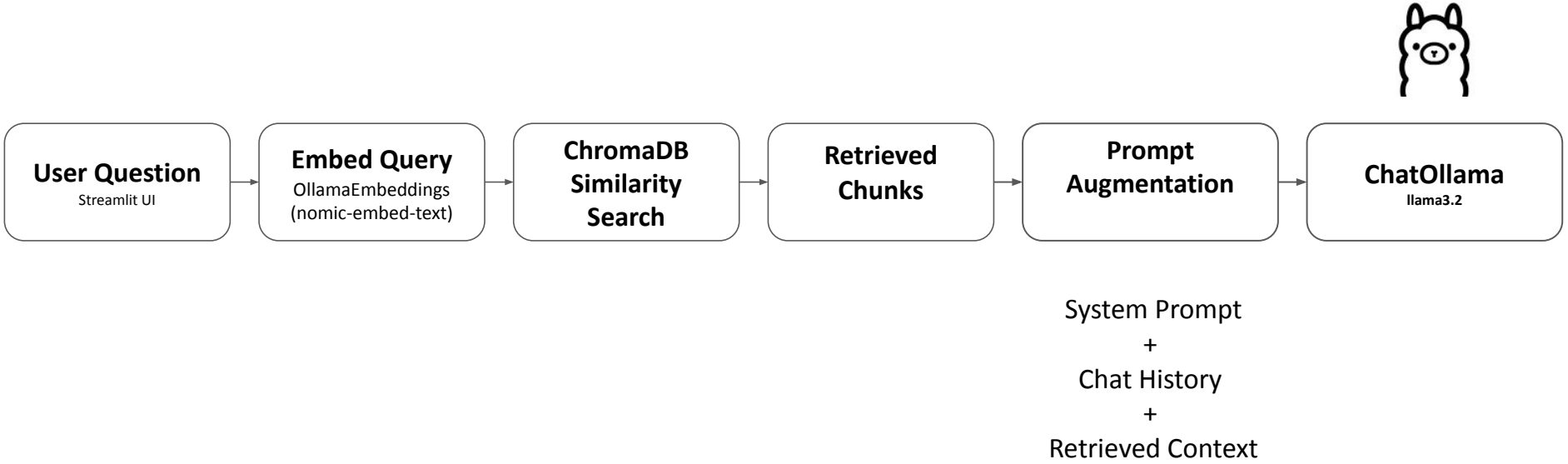
- Business knowledge must be dynamically integrated without retraining
- Source traceability is critical
- Cost-efficiency matters for scalability



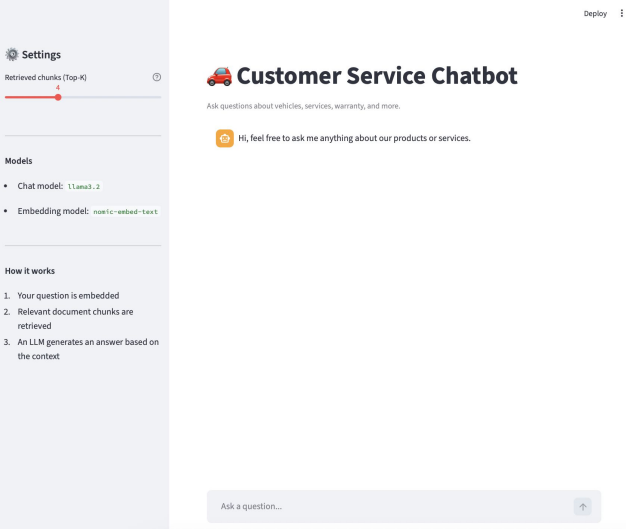
Customer Service RAG Chatbot Architecture Overview
Data Ingestion Pipeline



Customer Service RAG Chatbot Architecture Overview
RAG Chain

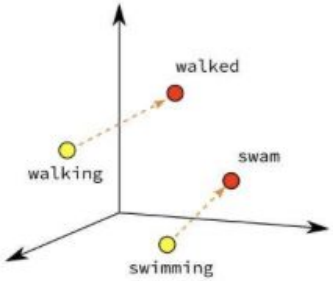


Customer Service RAG Chatbot Architecture Overview
Streamlit App



Key Features

- **Runs fully locally (Ollama + ChromaDB)**
  - No external API dependency
  - Suitable for prototype under CPU constraints
- **Chunking Strategy:RecursiveCharacterTextSplitter**
- **Nomic-embed-text**
  - Lightweight, runs locally via Ollama
  - Strong semantic similarity performance for its size
- **chunk_size=500, overlap=20**
  - 500 chars fits one topic per chunk without losing context
  - Overlap=20 prevents cutting mid-sentence at boundaries



Key Features

- **Retrieval Strategy**
  - Similarity-based Top-K search, configurable via UI
- **llama3.2**
  - 1–3B params
  - Runs on CPU
  - Good instruction-following
- **Chat History Handling**
  - Enables follow-up questions

Results - Verify Correct Information Retrieval

What is not covered under the standard warranty?

According to the provided context, specifically from doc_03_warranty, the following items are excluded from coverage under the Standard Warranty:

1. Normal wear items:
   - Brake pads
   - Tires
   - Wiper blades
   - Bulbs

Sources

🚗 Customer Service Chatbot

Ask questions about vehicles, services, warranty, and more.

Hello! I can help with vehicles, services, or warranty information.

Can I track my vehicle production?

You can track your vehicle's production status through our mobile app. According to the ordering process, you'll receive an order confirmation with an estimated delivery date after placing your order. Additionally, you can use the Online Configurator on our website or visit a dealer to configure your vehicle and save it with a code.

Once you've placed your order, you can track the production status via the mobile app. The typical production timeline is 6-10 weeks for standard configurations. You'll receive updates on the progress of your vehicle's production through the app.

Please note that you can also choose between dealer delivery or factory pickup experience during the ordering process.

Sources

- doc_02_service_maintenance
- doc_04_ordering_process

Ask a question...

Results - Verify Correct Information Retrieval

How long does fast charging take?

According to the provided context, fast charging for our electric vehicles takes approximately 30-35 minutes to charge from 10-80%.

Sources

- doc_05_electric_vehicles
- doc_03_warranty

I've sent an email about my issue. How long does email support take to respond?

According to our customer support channels, we have a 24-48 hour response time for emails through our online contact form. This means that you can expect a response within 24 hours, but it may take up to 48 hours in some cases.

Sources

- doc_06_customer_support
- doc_01_vehicle_features

Results - Verify Correct Information Retrieval

What happens if my battery capacity drops below 70%?

According to the provided context, specifically from doc_03_warranty under Electric Vehicle Battery Warranty, if your battery capacity drops below 70%, it is covered by the warranty. This means that you will receive coverage for battery cell defects and high-voltage components related to this issue. The warranty duration for this coverage is 8 years or 160,000 km (whichever comes first).

Sources

- doc_03_warranty
- doc_05_electric_vehicles

Can I change my configuration after production starts?

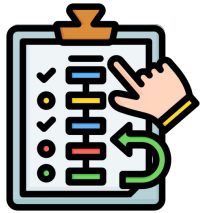
According to doc_01_vehicle_features, customers can save and share their configurations online. A completed configuration can be sent directly to a dealer for a personalized offer. Configuration changes are possible until the vehicle enters production.

In other words, yes, you can make changes to your configuration before the vehicle is produced. However, once production starts, further changes may not be allowed.

Sources

- doc_01_vehicle_features
- doc_04_ordering_process

Results - Follow-up Questions



How long is the EV battery warranty?

According to the provided context, the Electric Vehicle Battery Warranty has a duration of 8 years or 160,000 km (whichever comes first).

> Sources

What happens if capacity drops below 70%?

According to the provided context, if the capacity of the Electric Vehicle Battery drops below 70%, it is covered under the Electric Vehicle Battery Warranty. This warranty covers battery cell defects and high-voltage components.

> Sources

How long does vehicle production take?

According to the provided context, the typical production timeline for our electric vehicles is 6-10 weeks for standard configurations.

> Sources

Can I modify my order during that time?

According to the provided context, configuration changes are possible until the vehicle enters production. The exact duration of this period is not specified in the context. However, it's mentioned that once a vehicle enters production, configuration changes may be subject to certain limitations or restrictions.

It's recommended to contact our Customer Support channels (Customer Hotline or Online Support) for more information on modifying your order during the pre-production phase.

> Sources

Results - Follow-up Questions

🚗 Customer Service Chatbot

Ask questions about vehicles, services, warranty, and more.

Welcome! I'm your AI Assistant. What would you like to know?

Hi, what is included in standard warranty option?

The standard warranty option includes 2 years of unlimited mileage coverage for all manufacturing defects in materials and workmanship. This means that any issues related to the vehicle's build quality will be covered during this period. However, normal wear items such as brake pads, tires, wiper blades, and bulbs are excluded from this coverage.

> Sources

Are tires covered?

According to the standard warranty option, normal wear items such as tires are excluded from coverage. This means that tire-related issues or replacements are not included in the 2-year unlimited mileage warranty period.

> Sources

data > knowledge_base > doc_03_warranty.txt

```
Title: Warranty Coverage
Category: Warranty & Guarantees

We provide comprehensive warranty coverage for all new vehicles:

Standard Warranty:
- Duration: 2 years unlimited mileage
- Coverage: All manufacturing defects in materials and workmanship
- Exclusions: Normal wear items (brake pads, tires, wiper blades, bulbs)
```

Hallucination Test

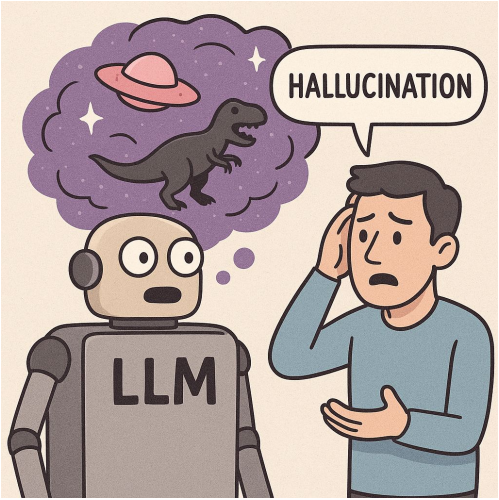
🚗 Customer Service Chatbot

Ask questions about vehicles, services, warranty, and more.

Welcome! I'm your AI Assistant. What would you like to know?

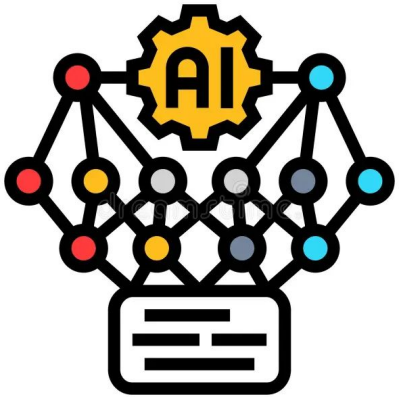
Do you offer Level 4 autonomous driving?

Sorry, I don't have information about this question. The provided context does not mention anything related to Level 4 autonomous driving. It discusses vehicle features, service intervals, ordering and production processes, customer support options, and complaint procedures, but there is no mention of autonomous driving capabilities or features.



Business Impact

- Reduced customer support workload
- Faster response times
- Consistent information delivery
- Improved digital customer journey
- Scalable knowledge interface
- Explainable solution
- Reduced hallucination



Limitations and Future Work

- **Limitations**
  - Retrieval quality depends on embedding performance
  - Slow response: Local CPU, no GPU
  - Small model: llama3.2 can hallucinate outside the given context
  - Static knowledge base: No real-time updates, manual re-ingestion required
- **Future Work**
  - Better models for production quality
  - Hybrid search (BM25 + vector search)
  - Stronger embedding model
  - Automatic evaluation framework (Precision@K, Recall@K)
  - Scalable vector store: ChromaDB → Pinecone / Weaviate
  - Hallucination detection metrics
  - Real-time document updates

Summary

What we built:

- A local RAG chatbot that answers customer questions based on an automotive knowledge base
  - End to end, from document ingestion to chat interface.

Key takeaway: RAG is the right approach here

- Grounded in real documents, transparent sources, easy to update

Thank you!

Github Repo:

https://github.com/aylinirem kara/Customer-Service-RAG-Chatbot-Local-AI-Prototype